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A Multi-Stage Post-Training Framework for Domain-Specific Language Models in Fault Diagnosis

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Abstract

The rapid advancement of large language models (LLMs) has created new opportunities for intelligent fault diagnosis, particularly in complex industrial systems, such as heating, ventilation, and air conditioning (HVAC) in urban rail transit. Although LLMs have shown strong general reasoning capabilities, adapting them to domain-specific fault diagnosis tasks remains challenging. This is particularly true for textual maintenance records, where sparse and brief entries can only provide limited context for effective knowledge adaptation. To address this challenge, we propose a multi-stage post-training framework based on LLMs. The framework consists of three components: (1) data augmentation via retrieval-augmented generation (RAG) to enrich brief maintenance records with domain knowledge and reasoning traces; (2) supervised fine-tuning (SFT) for domain-specific adaptation; and (3) reinforcement learning with group relative policy optimization (GRPO), using a task-specific reward that separately evaluates root-cause identification and maintenance action recommendation. The framework is applied to a real-world textual HVAC fault dataset derived from Ningbo Rail Transit, covering 33 equipment categories and over 100 fault types. With Qwen3-0.6B as the base model, the proposed method significantly improves diagnostic accuracy and reasoning quality, achieving a 97% increase in model-based diagnostic accuracy (from 0.323 to 0.635) and a 48% improvement in human expert evaluation scores (from 0.509 to 0.754). Moreover, conventional machine learning baselines, such as Support Vector Machine (SVM), Random Forest (RF), and Multi-Layer Perceptron (MLP), achieve accuracies below 0.4 in this task, further highlighting the superiority of the proposed LLM-based framework. These results indicate that the proposed multi-stage post-training framework effectively improves LLM performance on real-world text-based fault diagnosis. It provides a practical and extensible solution for intelligent maintenance decision support in complex electromechanical systems.

Keywords: fault diagnosis; HVAC; large language models; reinforcement learning; urban rail transit

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1. Introduction

Fault diagnosis in real-world maintenance often involves complex fault patterns, varying operating conditions, and unstructured maintenance records [1,2]. Conventional

knowledge-based and data-driven methods typically rely on explicit physical models or structured sensor data [3–6], making it difficult to fully exploit textual information such as fault descriptions, operational observations, and repair actions. Recent advances in large language models (LLMs) provide an alternative by enabling semantic understanding and contextual reasoning over such maintenance text [7–11]. Existing studies further suggest that LLM-based methods can complement conventional approaches by incorporating maintenance knowledge and contextual information into the diagnostic process [12,13].

These methodological advances have the potential to significantly enhance fault diagnosis for heating, ventilation, and air conditioning (HVAC) systems [14–16]. HVAC equipment failure not only compromises system reliability but also leads to additional energy losses [17–19], while timely troubleshooting is estimated to reduce energy consumption by 10–30% [20]. In urban rail transit (URT) systems, HVAC systems account for approximately 25–35% of total station energy consumption [21], and the complexity and diversity of HVAC failures tend to increase with operating mileage and service time.

To better reflect operational realities, we constructed a dataset based on textual records of HVAC maintenance and repair activities collected from Ningbo Rail Transit. This dataset spans five metro lines and 4.5 years, involving 33 equipment categories and more than 100 distinct fault types. The records, initiated by station personnel and subsequently handled by maintenance staff, are typically brief and often lack detailed descriptions. For example, a record documents the issue as "an office AC unit failed to provide heating". The maintenance personnel then appended the diagnosed cause, "dust accumulation on the filter", along with the corrective action, "cleaning the filter". Such brevity leads to insufficient contextual information. Moreover, a single fault phenomenon may stem from multiple underlying causes, further complicating reliable diagnosis from short texts.

Given these challenges, existing approaches often exhibit notable limitations. Knowledge-driven frameworks require laborious physical models and are difficult to scale to large HVAC systems with various operating states [5,6]. On the other hand, data-driven approaches are often designed for structured sensor and time-series data [22]; they are less effective when applied to corpora dominated by natural-language maintenance logs [6]. We use classical machine learning baselines, including Support Vector Machine (SVM), Random Forest (RF), and Multi-Layer Perceptron (MLP) [23–26], which are widely adopted in fault diagnosis tasks. Although LLMs offer a promising alternative for handling textual records, current applications of LLMs to engineering-scale diagnosis are still in their early stages [7–9]. Key challenges remain: bridging the gap between general pre-training and domain-specific maintenance texts, and reasoning effectively from brief fault descriptions.

To address these issues, we propose a multi-stage post-training framework based on LLMs. Specifically, we employ retrieval-augmented generation (RAG) [27,28] to enrich the dataset with Chain-of-Thought [29] reasoning, followed by supervised fine-tuning and Group Relative Policy Optimization (GRPO) [30], a reinforcement learning (RL) strategy for domain adaptation [11]. In the RL stage, the model is further optimized using a reward signal tailored to the requirements of HVAC fault diagnosis. This allows the post-training process to account for both diagnostic accuracy and the quality of recommended maintenance actions. The experimental results show that the proposed approach achieves effective fault diagnosis, while evaluations by domain experts indicate a high level of satisfaction with the generated responses.

1.1. Contributions

This study makes the following contributions to the field of HVAC fault diagnosis:

- We construct and analyze a large-scale textual HVAC maintenance dataset comprising 4195 real operational records collected over 4.5 years from Ningbo Rail Transit.

- We develop a post-training framework that uses RAG to augment the training data with domain-specific knowledge and Chain-of-Thought reasoning, followed by supervised fine-tuning and GRPO-based reinforcement learning.
- We design a task-specific reward function for GRPO that separately evaluates root-cause identification and maintenance action recommendation, reflecting the two main steps of HVAC troubleshooting.
- The experimental results demonstrate that our proposed framework achieves the best performance in both model- and human-based evaluations, with scores of 0.768 and 0.754 on the test set, respectively, significantly outperforming baseline models such as traditional machine learning models and untrained Qwen models.

1.2. Outline

The rest of this paper is structured as follows: Section 2 reviews the related work. Section 3 presents a real-world HVAC fault and maintenance dataset collected from Ningbo Rail Transit. Section 4 describes the proposed multi-stage post-training framework. Section 5 outlines experimental settings, evaluates model performance, analyzes the results, and discusses limitations. Finally, Section 6 summarizes the main findings and discusses their practical implications.

2. Related Work

This section primarily covers HVAC fault diagnosis algorithms and the application of LLMs in fault diagnosis. Research on applying LLMs to HVAC systems is discussed in Section 2.2.

2.1. Fault Diagnosis in HVAC Systems

HVAC fault diagnosis has shifted toward two dominant paradigms, i.e., data-driven and knowledge-based approaches [4–6]. Data-driven methods, based on machine learning techniques such as decision trees [31] and Support Vector Machines [23], have been widely applied to this task.

More recently, neural network-based methods have attracted considerable attention. For example, a hybrid framework that integrates a basic neural network with an auxiliary network has been proposed for anomaly detection in HVAC [32]. In addition, convolutional neural networks with softsign activation have demonstrated strong performance on the RP-1043 dataset [33], while pruning algorithms have been employed to reduce model parameters without sacrificing competitiveness [34]. In contrast, knowledge-driven methods rely on domain expertise for model construction, such as employing Bayesian networks that incorporate prior knowledge to support fault diagnosis [35].

2.2. Large Language Models in Fault Diagnosis Applications

The application of LLMs to fault diagnosis has emerged as a growing research focus, with some studies exploring their use in the HVAC domain. Fine-tuning strategies that integrate both fault and fault-free labeled data have been shown to be effective on the Air Handling Unit (AHU) fault dataset from the RP-1312 project [36]. LLMs have also been applied to HVAC time-series data, where labeled datasets are used for fault classification [37]. Furthermore, evaluations of domain-specific knowledge acquisition indicate that, although LLMs already outperform half of the human participants, considerable potential for broader application remains [38].

Beyond HVAC, fault diagnosis in other domains has also advanced considerably with the integration of LLMs [9,39–42]. Recent studies have increasingly focused on enhancing LLM-based fault diagnosis, where the authors of [43] proposed the FD-LLM framework for complex equipment diagnosis, demonstrating strong industrial applicability. For instance,

fine-tuned LLMs have been validated on the TEP dataset for efficient system fault diagnosis, confirming their practical effectiveness [9]. In the bearing fault domain, LLMs have been utilized to transform signals into textual representations, enabling analysis of quantized signal characteristics and improving data generalization [39].

For machinery maintenance, an LLM with Chain-of-Thought reasoning has been used to enhance data augmentation for few-shot soft contrastive learning in fault diagnosis, effectively improving feature consistency and generalization capability [40]. For cross-domain fault diagnosis with class mismatch, Chen et al. [44] proposed a prompt-guided disentanglement and fusion framework that separates domain-specific and fault-related features to improve cross-domain generalization. A knowledge graph of assembly processes has been integrated with an LLM to support fault diagnosis, demonstrating promising performance in real-world applications [41]. In addition, LLM-based prognostic methods have emerged, where the authors of [45] proposed a framework aligning signal data with textual representations to support both fault diagnosis and predictive maintenance.

3. Dataset

The dataset used in this paper comprises comprehensive maintenance records of HVAC systems from Ningbo Rail Transit, covering five metro lines from December 2020 to May 2025. After preprocessing to eliminate false alarms, duplicates, and incomplete entries, a curated dataset of 4195 validated fault records was obtained.

Unlike prior studies based on structured sensor data or synthetic labels, this dataset captures the ambiguity, sparsity, and multi-causality of real maintenance records, which pose unique challenges for LLM-based reasoning.

Each entry is structured into seven key attributes:

- Time: Provides the timestamp of the fault occurrence with minute-level granularity.
- Line: Identifies the metro line where the fault occurred.
- Station: Identifies the specific station associated with the fault.
- Equipment Category: Classifies the type of HVAC equipment involved (e.g., multi-connected AC unit, chiller unit, air duct, and accessories).
- Fault Description: Documents the observed anomaly or alarm state reported by monitoring systems, operators, or other station personnel (e.g., abnormal fan vibration, AC unit fails to start remotely).
- Root Cause: Specifies the root-cause diagnosis as determined by maintenance staff (e.g., filter clogging, wiring looseness, power outage).
- Action: Records the corrective measures undertaken (e.g., filter cleaning, wire fastening, or equipment reset).

As shown in Figure 1, maintenance and repair tasks for HVAC systems encompass both outdoor equipment, such as cooling towers, and indoor equipment, including chillers and associated electrical ducting. This reflects the diverse operational environments and equipment types represented within the dataset.

3.1. Visualization

The dataset includes 33 equipment categories across HVAC subsystems, with the most frequent ones shown in Figure 2a. Notably, the top five categories alone account for nearly half of all recorded cases, highlighting their critical role in metro station HVAC infrastructure.

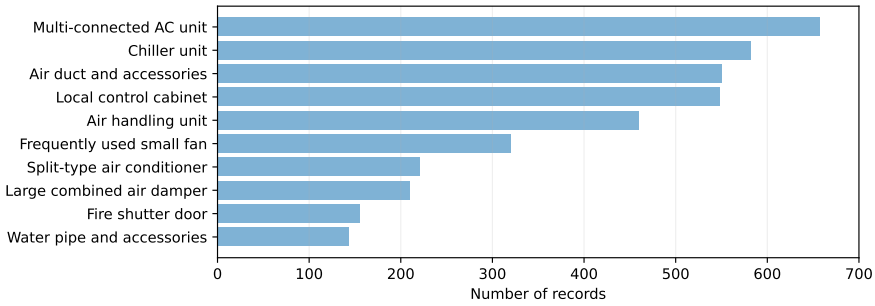
Furthermore, the dataset captures a wide spectrum of fault causes, spanning mechanical, electrical, and operational issues. As illustrated in Figure 2b, the most common diagnoses include power outages, improperly positioned air dampers, clogged air filters, faulty actuators, and bearing failures.



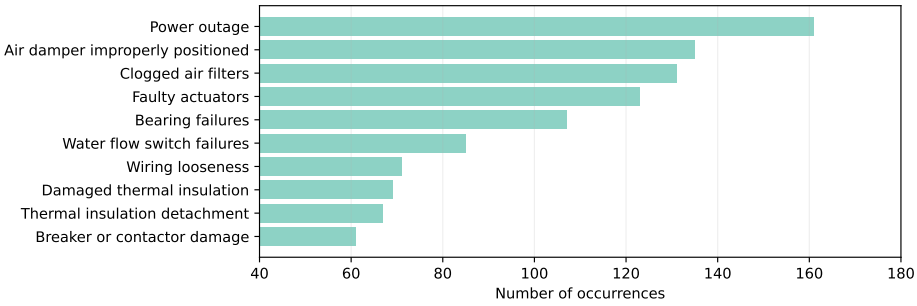
(**a**) Inspection of the cooling tower.

(**b**) Inspection of electrical conduits within the suspended ceiling.

Figure 1. Typical maintenance operations for HVAC systems in urban rail transit stations.



(**a**) Top 10 equipment categories by fault records.



(**b**) Top 10 fault causes by occurrence frequency.

Figure 2. Visualization of the maintenance dataset.

Based on domain knowledge and maintenance practices, these fault causes can be grouped into two broad types: The first consists of routine maintenance issues, such as clogged filters and deteriorated insulation, which primarily reflect the ongoing service demands of metro HVAC operations. The second involves component-level failures, including defective bearings, breakers, and actuators, which pose more serious challenges to system reliability and longevity. Taken together, these failure types capture both the short-term maintenance workload and the long-term operational risks inherent in metro HVAC systems.

3.2. Data Splitting

The data were chronologically split to avoid leakage: 2932 records from December 2020 to May 2024 for training, and 1263 records from June 2024 to May 2025 for independent testing.

4. Methodology

The overall methodological framework is illustrated in Figure 3 and summarized in Algorithm 1, consisting of three main components: an RAG-based data augmentation algorithm, followed by two post-training strategies, namely, SFT and RL. The framework consists of three stages: retrieval, distillation, and optimization. In Stage 1, we construct an augmented training set via RAG. Chain-of-Thought (CoT) prompting enriches each sample with fault analysis and diagnostic reasoning, and the resulting corpus is quality-filtered for subsequent use. In Stages 2 and 3, we perform post-training: SFT provides a cold start, followed by GRPO-based RL to strengthen reasoning. This sequence enables the language model to gradually acquire domain-specific knowledge and improve diagnostic accuracy.

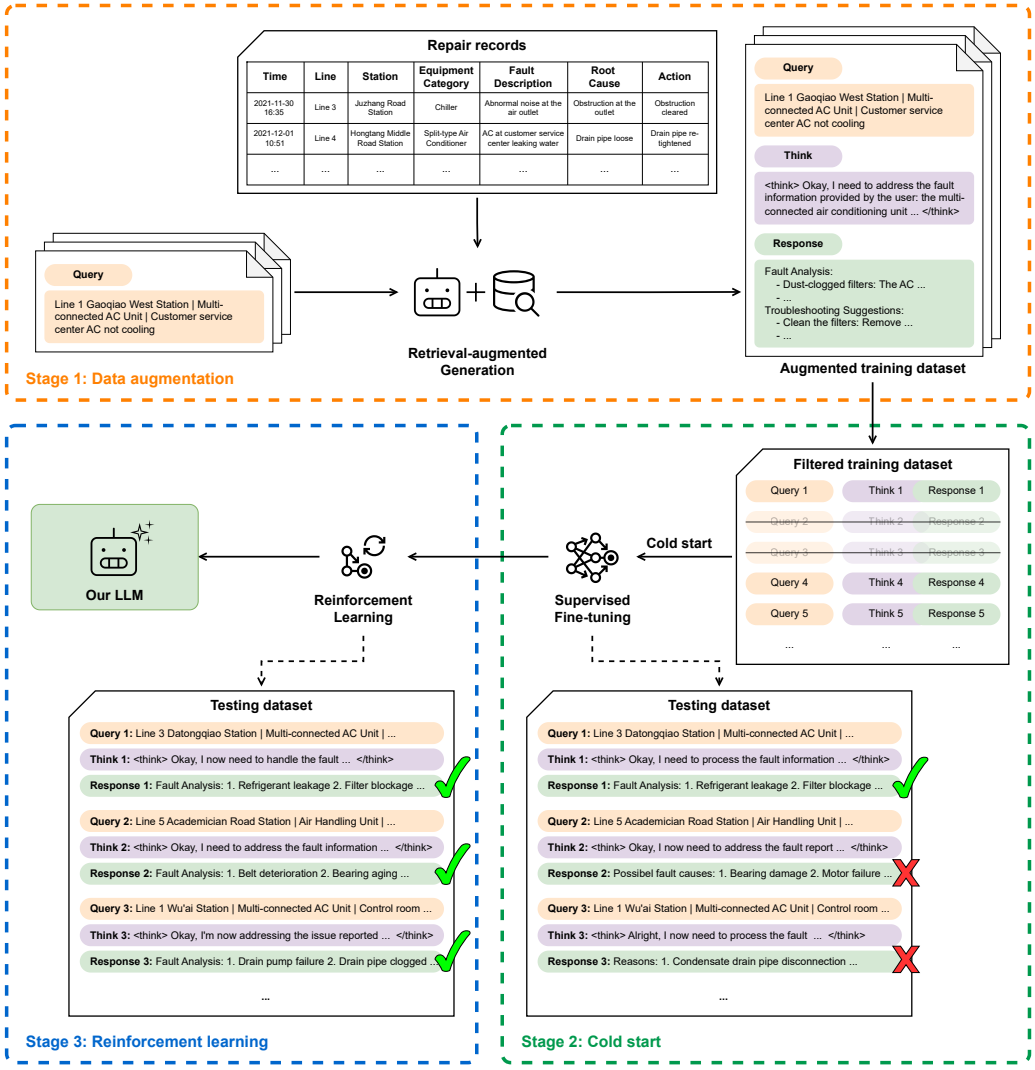


Figure 3. The proposed post-training framework. Stage 1 augments and filters training data via RAG; Stages 2 and 3 apply SFT and RL, respectively, enabling the language model to acquire domain-specific diagnostic knowledge and reasoning capability.

We formalize the diagnosis task as follows: Given a structured input x, the model must generate two sets of outputs: a set of candidate root causes $Y_c = \{c_1, \ldots, c_m\}$, and a set of candidate maintenance actions $Y_a = \{a_1, \ldots, a_n\}$. This setting combines aspects of short-text generation and multi-label classification. In our evaluation, the model output is decomposed into a cause component y_c and an action component y_a; each component may contain multiple candidate items.

Given the constraints of GPU memory and model size, Qwen3-0.6B [11] was adopted as the baseline, serving as the foundation for all subsequent model refinement.

Algorithm 1 Multi-Stage Post-Training for HVAC Fault Diagnosis

Require: Maintenance dataset D, pre-trained LLM π_0
Ensure: Optimized diagnostic model π^*
Stage 1: RAG-based data augmentation
for each record $d \in D$ **do**
  Retrieve relevant cases and generate reasoning traces.
end for
Construct and filter the augmented dataset D'.
Stage 2: Supervised fine-tuning
Fine-tune π_0 on D' to obtain π_{SFT}.
Stage 3: GRPO-based reinforcement learning
Initialize $\pi_\theta \leftarrow \pi_{\text{SFT}}$.
for each training step **do**
  Generate a response group $\{o^{(i)}\}$ for each query.
  Compute the composite reward in Equation (5).
  Compute group-relative advantages and update π_θ.
end for
return $\pi^* \leftarrow \pi_\theta$

4.1. Benchmarking the Original Model

We evaluated the performance of the original Qwen3-0.6B on HVAC system fault diagnosis using operational data from Ningbo Rail Transit. This evaluation aimed to preliminarily assess the fault identification capability of the Qwen3 model and to design an appropriate prompting strategy for subsequent dataset construction and post-training.

Prompt engineering has become an essential technique for enhancing the capabilities of LLMs [46]. The prompt framework in this work consists of three components: the system prompt (or instructions), the context, and the user prompt, as shown in Figure 4. The system prompt defines an HVAC fault maintenance expert by providing relevant background information and specifying the response requirements. The expert is then expected to identify potential fault causes and recommend appropriate corrective actions. The context provides supplementary technical information, including historical dialogue records and retrieved content when RAG is enabled. The user prompt conveys the real-time fault description from operators. We transform operating data into inputs for LLMs using the template illustrated in Figure 4; the template contains four fields: Line, Station, Equipment Category, and Fault Description.

Figure 5 presents a specific HVAC maintenance case, where the integrated monitoring system issued an alarm due to the unexpected shutdown of the chiller unit. The model-generated response is divided into two components: the "think" part provides an analysis of the problem, and the "answer" part generates possible causes and recommended actions. While this response is well structured, the model fails to find the root cause due to the lack of relevant domain-specific knowledge.

It should be noted that the prompt framework and system prompt shown in Figure 4 are consistently adopted as the default configuration. To maintain clarity, this system prompt is omitted in the following dialogue examples.

System

You are a seasoned maintenance engineer for electromechanical equipment in rail transit, with a solid theoretical foundation and extensive field experience. Please always respond in this capacity, maintaining professional, practical, and clear language.

Based on the user's fault input format "{Line}{Station}|{Equipment Category}|{Fault Description}", analyze the possible causes of the fault and provide professional judgments and recommendations. Structure your response as follows:

1. **Possible Causes**: List and analyze potential fault reasons in bullet points.
2. **Recommended Actions**: Propose and explain troubleshooting solutions in bullet points.

Context

{*Conversation history*} and {*Reference information (from RAG, if any)*}

User

{*Line*}{*Station*} | {*Equipment Category*} | {*Fault Description*}

Figure 4. Prompt Framework: (1) system prompt defines the model's role as an HVAC maintenance expert; (2) context provides supplementary fault details; (3) user prompt submits the specific diagnostic query.

User

Line 3 Sakura Park Station | Chiller unit | Integrated monitoring system alarm: chiller unit abnormal shutdown

Assistant

<think> Okay, I now need to process the fault information provided by the user: Abnormal shutdown alarm of the chiller at Sakura Park Station on Line 3. First, I must understand the possible causes of this fault. The user requested a bullet-point list of causes and handling suggestions, so I need to carefully analyze potential factors.

First, abnormal shutdown alarms of chillers can be caused by various reasons. Common ones include overload, temperature abnormalities, power supply faults, system malfunctions, or abnormal conditions. I need to consider each of these possibilities one by one. For example, temperature abnormalities might be due to excessively high or low cooling water temperatures, preventing the chiller from operating normally. Power issues could involve unstable input voltage or power outages, affecting the chiller's normal operation. System faults might involve problems with the control module or abnormal sensor data. Additionally, it is necessary to consider whether issues with other interconnected devices, such as the emergency braking system triggering a shutdown, are causing the problem.

Next is the handling suggestions section. Specific solutions need to be proposed for each possible cause. For instance, check whether the power supply voltage is stable and ensure the cooling water temperature is within the normal range. If it is a control module issue, replacement or inspection of related components may be required. For abnormal sensor data, calibration or replacement of the sensor might be necessary. At the same time, check whether the emergency braking system is triggered and ensure no other interconnected devices are causing the shutdown.

It is also important to consider the user's potential deeper needs, such as how to quickly locate the problem or how to prevent similar situations. Suggestions for regular maintenance and inspections, as well as monitoring system status, may be necessary. Ensure the response is clearly structured, with distinct sections and sub-points, so the user can easily understand and implement the suggestions. </think>

1. **Possible Causes**:
   - **Temperature Abnormalities**: Excessively high or low cooling water temperatures during chiller operation may cause the system to misinterpret the shutdown status.
   - **Power Supply Fault**: Fluctuations in input voltage or power outages can prevent the chiller from starting, affecting the alarm function.
   - **Control Module Abnormalities**: Abnormal input or output of control signals may cause the system to misinterpret the shutdown status.
   - **System Interconnection Fault**: Misactivation of the emergency braking system triggering a shutdown, leading to an abnormal shutdown alarm for the chiller.
2. **Recommended Actions**:
   - **Power Supply Check**: Ensure the chiller's input voltage is stable within the normal range (e.g., 220V ± 15\%), and rule out faults caused by power outages or voltage abnormalities.
   - **Temperature Monitoring**: Check whether the cooling water temperature sensor is functioning properly, and calibrate or replace the sensor to ensure accurate temperature data.
   - **Control Module Check**: Inspect whether the control signal input and output are normal, and troubleshoot any faults or delays in the control module.
   - **Emergency Braking Interconnection**: Confirm that the emergency braking system has not been misactivated, and check whether other equipment (e.g., train signal systems) is interconnected with the chiller to ensure no misoperations are causing the shutdown.
   - **Regular Maintenance**: Regularly inspect the chiller and related equipment to prevent shutdown alarms caused by aging or abnormalities.

Figure 5. Example of HVAC fault diagnosis using the original Qwen3-0.6B model on the test set. The root cause of the fault is water flow switch failure, and the maintenance personnel resolve the issue by replacing the water flow switch. This response does not provide the correct diagnosis.

4.2. Benchmarking the Model with Retrieval-Augmented Generation

RAG [47] enhances LLMs by retrieving relevant information from an external knowledge base prior to response generation, thereby improving accuracy and mitigating hallucinations [27]. In our setting, fault descriptions are first standardized into a unified input format (Figure 4), which allows an embedding model to retrieve the top-k semantically similar cases from the training dataset. These retrieved examples provide question-specific contextual knowledge, enabling the model to generate more accurate and grounded responses.

Figure 6 presents the result of RAG in an HVAC maintenance case, using the same example as in Figure 5 for comparison. We observe that, with RAG, the model retrieves semantically similar cases from the training dataset during the "think" phase. For instance, given the original query "Integrated monitoring system alarm: chiller unit abnormal shutdown", related cases such as "Chiller unit abnormal shutdown alarm" at Gaotangqiao Station and "Chiller unit shutdown alarm" at Zhonggongmiao Station are retrieved. Based on these historical cases, the model is able to identify underlying issues and recommend corresponding corrective actions in the "answer" phase. This demonstrates that RAG enhances the model's comprehension and improves its performance by incorporating relevant contextual information. However, although the historical records mention issues with water flow, they do not identify a faulty water flow switch as the root cause. Therefore, while the RAG provides the crucial information about insufficient water flow, its response is not entirely accurate.

Data Augmentation Using RAG

LLMs typically require large and diverse datasets to achieve good performance. Insufficient or low-quality training data may lead to poor generalization, limiting the ability to handle complex tasks. In our maintenance dataset, the original records contain only isolated causes and solutions without any explicit reasoning process. However, for a single HVAC diagnostic query, multiple potential causes may exist, and identifying the true cause often requires a process of reasoning. These limitations make the original dataset less suitable for fine-tuning or other downstream tasks.

RAG has been shown to improve both data quality and reasoning performance [27,48]. For each training record, we form a query using its structured attributes and retrieve the top-k semantically similar historical records ($k = 10$). Only samples strictly within the training period are used as retrieval candidates, which avoids future information leakage and ensures that the augmented CoT reflects realistic operational context. Additionally, the target record is excluded from the retrieval candidates, preventing its ground-truth cause and action from being reused during augmentation. By enriching brief maintenance records with reasoning chains and relevant fault causes, RAG provides more informative training data for the subsequent model post-training.

4.3. Post-Training Strategy

We adopt a post-training approach that enables the thinking (or reasoning) mode. First, we focus on initializing the model's reasoning ability for repair problems through cold-start fine-tuning with long CoT. Subsequently, we use RL to develop strong reasoning capabilities.

4.3.1. Supervised Fine-Tuning as the Cold Start

Given that the pre-trained model lacks sufficient domain knowledge in HVAC maintenance, this stage aims to initialize its reasoning ability and embed a fundamental understanding of fault root causes along with corresponding repair strategies. It also serves as an intermediate distillation stage between retrieval-augmented data construction and rein-

forcement learning. We use the CoT-augmented training data generated through the aforementioned RAG process as the foundation for this stage. By distilling retrieval-augmented knowledge into the model parameters, this design decouples the retrieval step from the subsequent RL process. As a result, RL can proceed without database access, significantly reducing overall training costs.

User

Line 3 Sakura Park Station | Chiller unit | Integrated monitoring system alarm: chiller unit abnormal shutdown

Assistant

<think> Okay, I now need to handle the user's inquiry regarding the abnormal shutdown alarm of the chiller unit at Sakura Park Station on Line 3. First, I should review previously provided cases to see if there are similar fault scenarios. The user's current equipment is a chiller unit, and the fault symptom is an abnormal shutdown displayed on the integrated monitoring system. In the cases provided by the user, similar faults occurred at Gaotangqiao Station on Line 3 and Zhonggongmiao Station on Line 5, such as insufficient condenser water flow or excessively high compressor refrigerant outlet temperature. The user likely expects a detailed analysis and handling suggestions for this situation.

Next, I need to identify the cause of the fault. Based on previous cases, abnormal shutdowns of chiller units can be caused by various factors, such as insufficient water flow, excessive pressure difference, condenser scaling, or cooling water interruption. These causes need to be categorized to ensure each has corresponding handling measures. At the same time, the suggestions should be logically clear for the user to understand and implement.

The user's unspoken deeper need may be to understand how to prevent similar issues or how to quickly locate the source of the fault. Therefore, the response should not only list the causes and solutions but also provide preventive measures or follow-up inspection suggestions to help the user avoid similar faults in the future.

It is necessary to check if any fault causes or handling methods have been omitted to ensure the information is comprehensive and practical. Meanwhile, the response should be clearly structured, listing fault causes and handling suggestions in bullet points, using professional terminology while avoiding excessive complexity to ensure the user can easily understand. </think>

1. **Possible Causes**:
   - **Insufficient Water Flow**: Inadequate water supply to the condenser or evaporator, preventing the system from maintaining normal operation.
   - **Excessive Pressure Difference**: Abnormal increase in condensation pressure in the condenser or evaporator, potentially causing overheating or pressure loss of control.
   - **Condenser Scaling**: Reduced heat exchange efficiency due to scaling during water flow, affecting exhaust pressure.
   - **Excessively High Refrigerant Outlet Temperature**: Abnormal increase in compressor refrigerant outlet temperature, potentially causing system overheating or fault alarms.
   - **Cooling Water Interruption**: Interruption in the cooling system preventing the refrigerant from circulating normally, affecting overall operation.
2. **Recommended Actions**:
   - **Check the Water System**: Ensure water pumps, pipelines, filters, and other water supply equipment are functioning properly. Increase water supply if necessary.
   - **Adjust Pressure**: If pressure is abnormal, restore system balance by adjusting physical parameters (e.g., valve opening) of the condenser or evaporator.
   - **Clean the Condenser**: If scaling causes performance degradation, clean the condenser surface and replace filters if necessary.
   - **Check the Refrigerant System**: Ensure the refrigerant pump is operating normally and the refrigerant outlet temperature is within a safe range.
   - **Reset the Alarm**: If the alarm is due to pressure or temperature abnormalities, prioritize resetting the system and observe for a period before troubleshooting.
   - **Record and Analyze**: Document the time of the fault, environmental parameters, and handling process to facilitate future fault diagnosis.

Figure 6. Example of HVAC fault diagnosis using the Qwen3-0.6B RAG model on the test set. The root cause of the fault is water flow switch failure, and the maintenance personnel resolve the issue by replacing the water flow switch. The green text in the "think" section indicates that the model successfully retrieves and uses related cases. The model mentions issues with the water system but does not identify the water flow switch as the root cause.

Although the CoT generated by RAG may contain noise, it generally preserves valid reasoning structures and reasonable causal relationships. These weakly supervised signals are sufficient to initialize the model's reasoning capability, while subsequent RL further refines these patterns.

Following the guidelines in [11], we filter out queries that could be correctly answered without CoT reasoning, thereby allowing the model to focus on more complex questions.

In particular, we exclude straightforward, solvable cases in which the fault description, root cause, and recommended action are nearly identical. For example, when the description states that the insulation cotton is damaged, the identified cause is simply "damaged insulation cotton", and the prescribed action is to repair or replace the insulation cotton. Such instances do not require multi-step reasoning or deductive analysis.

To support effective RL in the subsequent phase, we reduce both the number of training samples and training iterations, while ensuring that the model is exposed to all essential failure types and their corresponding solutions. This strategy prevents excessive enhancement of reasoning capabilities at this early stage.

In addition, current fine-tuning approaches are generally categorized into parameter-efficient and full-parameter schemes. We adopt full-parameter fine-tuning. Although parameter-efficient methods such as low-rank adaptation (LoRA) [49] can approximate weight updates using the product of two low-rank matrices to significantly reduce computational and memory overhead, their performance on complex tasks, such as mathematical reasoning, is often inferior to that of full-parameter fine-tuning [50]. Therefore, full-parameter fine-tuning is employed to achieve superior task performance.

4.3.2. Group Relative Policy Optimization and Semantic Reward

GRPO [30] significantly enhances the reasoning ability of LLMs. GRPO is based on proximal policy optimization (PPO) [51] and is widely used in the fine-tuning phase of LLMs. It estimates the advantage based on the relative performance within a group of actions, thereby eliminating the reliance on the value function to reduce the demand for GPU memory. Inspired by its success, we adopt the GRPO method to train reasoning models.

In LLM policy optimization, let the model policy be parameterized by θ. For a problem q drawn from a dataset $\mathcal{Q}$, a group of outputs $\{o^{(i)}\}_{i=1}^{n}$ is sampled from the old policy $\pi_{\theta\text{old}}$. A reward model then assigns scores $\{r^{(i)}(q)\}_{i=1}^{n}$ to these outputs. The training objective of GRPO [30] is defined as

$$\begin{aligned}\mathcal{G}_{\text{GRPO}}(\theta) &= \mathbb{E}_{q\sim\mathcal{Q},\{o^{(i)}\}_{i=1}^{n}} \frac{1}{n}\sum_{i=1}^{n}\frac{1}{|o^{(i)}|}\sum_{t=1}^{|o^{(i)}|}\Big\{\min\big[\rho_t^{(i)}A_t^{(i)},\\ &\qquad \text{clip}(\rho_t^{(i)},1-\epsilon,1+\epsilon)\,A_t^{(i)}\big]-\beta\mathbb{D}_{\text{KL}}\big(\pi_\theta\,\|\,\pi_{\text{ref}}\big)\Big\}, \qquad (1)\\ \rho_t^{(i)} &\equiv \rho_t^{(i)}(\theta)=\frac{\pi_\theta(o_t^{(i)}\mid q,o_{<t}^{(i)})}{\pi_{\theta\text{old}}(o_t^{(i)}\mid q,o_{<t}^{(i)})}, \qquad (2)\\ \mathbb{D}_{\text{KL}}\big(\pi_\theta\,\|\,\pi_{\text{ref}}\big) &= \frac{\pi_{\text{ref}}(o_t^{(i)}\mid q,o_{<t}^{(i)})}{\pi_\theta(o_t^{(i)}\mid q,o_{<t}^{(i)})}-\log\frac{\pi_{\text{ref}}(o_t^{(i)}\mid q,o_{<t}^{(i)})}{\pi_\theta(o_t^{(i)}\mid q,o_{<t}^{(i)})}-1, \qquad (3)\end{aligned}$$

where $\rho_t^{(i)}$ is a probability ratio that compares the current policy π_θ with the old policy $\pi_{\theta\text{old}}$, and ϵ controls the range of the clipped probability ratio. The KL divergence regularization between the trained policy and the reference policy is added to $\mathcal{G}$, and β regulates this penalty. The reference model π_{ref} is usually the initial SFT model. The advantage $A_t^{(i)}$ [30] in a response is calculated by normalizing the rewards r:

$$A_t^{(i)}=\frac{r^{(i)}-\text{mean}\big(\{r^{(i)}\}\big)}{\text{std}\big(\{r^{(i)}\}\big)}. \qquad (4)$$

The reward function is a key component of the RL stage because it determines which aspects of the generated diagnosis are reinforced. Rather than relying only on general

response quality, we design the reward to reflect the structure of HVAC troubleshooting. The overall reward is defined as

$$r = \alpha_1 r_{\text{format}} + \alpha_2 r_{\text{content}} + \alpha_3 r_{\text{semantic}}, \tag{5}$$

where r_{format} is the format score, which evaluates whether the model correctly switches between "think" and "answer" modes and uses the specified tags (i.e., `<think>` and `</think>`) to separate the thinking part from the response part. The content reward r_{content} assesses the comprehensiveness of the generated response, rewarding responses that provide multiple possible causes and corresponding corrective actions. The semantic reward r_{semantic} is further divided into two task-specific components: root-cause identification, and maintenance action recommendation. This decomposition separately evaluates whether the model identifies the underlying fault correctly and recommends an appropriate maintenance action, rather than assigning a single semantic score to the entire diagnosis. It is defined as

$$r_{\text{semantic}} = \mathbb{1}\{g(y_c, \hat{y}_c) \geq t\} + \mathbb{1}\{g(y_a, \hat{y}_a) \geq t\}, \tag{6}$$

where $\mathbb{1}$ denotes the indicator function, $g(\cdot) \in [0, 1]$ is a text reranking model [52–54] measuring semantic similarity between the generated response y and the ground truth $\hat{y}$, y_c and y_a represent the *cause* and *action* parts of the response y, respectively, and t is the similarity threshold. When multiple *cause* and *action* candidates are generated, the top five candidates are compared with the ground truth, and the one with the highest similarity is retained.

The format and content rewards, r_{format} and r_{content}, adopt a binary scoring scheme, assigning a score of 1 if the condition is satisfied and 0 otherwise. The hyperparameters α_1, α_2, and α_3 control the relative weights of the reward components. The reward weights reflect the operational priorities of HVAC diagnosis. Semantic accuracy is the primary determinant of diagnostic usefulness. Therefore, in our experiments, α_3 is set larger than α_1 and α_2. In contrast, r_{format} and r_{content} focus on structural correctness and completeness, which require lower weights to avoid overpowering semantic supervision. This weighting strategy is consistent with the RLHF literature [30], where semantic rewards typically dominate reasoning-oriented tasks. Further analysis of the individual contributions of the format, content, and semantic rewards, as well as their relative weights, will be considered in future work.

4.3.3. Benchmarking the Reinforcement Learning Model

Figure 7 presents the results of reinforcement learning for the same example in Figure 5. We observe that the model correctly identifies the underlying cause "Water flow switch failure" and provides an appropriate solution "Check the water flow switch: ... Replace if necessary". Compared to the original model, the RL-trained model exhibits a clear reasoning chain from potential causes to corresponding solutions, reflecting a more rigorous diagnostic process. Moreover, it outperforms the RAG model by more accurately identifying the underlying causes and generating correct solutions.

Notably, the RL-trained model tends to generate more concise "Possible Causes" and "Recommended Actions" than the other models. Here, we briefly analyze the possible reasons. To maximize the reward, the model may align its responses closely with the reference answers, which are inherently brief. In addition, a response-level length bias introduced by the $\frac{1}{|o^{(i)}|}$ term in Equation (2) assigns larger model updates to shorter but correct outputs [55]. Moreover, hardware constraints limit the maximum response length to 1024 tokens, further contributing to the overall brevity of the generated answers.

User

Line 3 Sakura Park Station | Chiller unit | Integrated monitoring system alarm: chiller unit abnormal shutdown

Assistant

<think> Okay, the user currently needs to address the abnormal shutdown alarm of the chiller unit at Sakura Park Station on Line 3. First, I need to review previous cases to see if there are any similar fault records. For example, in past cases, chiller unit alarms may have been caused by issues such as water flow switch failure, excessively high cooling water temperature, or excessively low oil pressure difference.

The current fault description provided by the user is ``the integrated monitoring system shows an abnormal shutdown alarm for the chiller unit,'' so the first step is to confirm the specific details of the alarm, such as whether there is a specific fault code or phenomenon. However, since the user did not provide these details, further inquiry may be needed. Based on previous cases, common causes of alarms typically include water flow switch failure, excessively high cooling water temperature, excessively low oil pressure difference, or insufficient cooling water flow.

Next, I need to analyze possible causes. For instance, a water flow switch failure may prevent the chiller unit from starting normally, triggering the alarm. Excessively high cooling water temperature may require checking the operation of the cooling tower to ensure the fan and water distributor are functioning properly. Excessively low oil pressure difference may necessitate inspecting the oil pressure sensor or checking for leaks in the pipelines. Additionally, insufficient cooling water flow may also prevent startup, requiring an inspection of the water supply system.

In terms of handling recommendations, the first step should be to confirm the specific circumstances of the alarm and then troubleshoot based on the possible causes one by one. For example, if the water flow switch is faulty, it may be necessary to check whether the switch is functioning properly or if there is any blockage or damage. If the issue is related to excessively high cooling water temperature, the relevant equipment should be inspected to ensure it is operating normally. At the same time, resetting the device is a common but important step to ensure the problem is resolved.

Additionally, the user may need to know how to contact professionals for further inspection or whether there is backup equipment that can be used as a substitute. It is also advisable to recommend regular maintenance and inspections to prevent similar faults from occurring. The response should be clearly structured, with causes and handling suggestions listed in bullet points to meet the user's requested format. </think>

1. **Possible Causes**:
   - Water flow switch failure preventing the chiller unit from starting, triggering the integrated monitoring alarm;
   - Excessively high cooling water temperature (e.g., fan failure or water distributor jam);
   - Excessively low oil pressure difference (oil pressure sensor or pipeline leakage);
   - Insufficient cooling water flow (water pump or pipeline failure).
2. **Recommended Actions**:
   - **Check the water flow switch**: Confirm whether the switch is functioning properly and check for blockages or damage. Replace if necessary.
   - **Troubleshoot the cooling water system**: Inspect the fan operation status and water distributor rotation to ensure normal cooling water circulation.
   - **Check oil pressure and pressure differences**: Verify whether the oil pressure sensor is functioning properly and investigate any abnormal oil pressure differences.
   - **Supplement or adjust water flow**: If the flow is insufficient, inspect the water pump, pipelines, or filters to ensure a stable water supply.
   - **Reset the equipment**: If the fault is reset-related, follow the equipment manual to reset and observe whether normal operation resumes.

Figure 7. Example of HVAC fault diagnosis using the GRPO-based model on the test set. The root cause of the fault is water flow switch failure, and the maintenance personnel resolve the issue by replacing the water flow switch. The model successfully identified the correct fault cause through post-training, as highlighted in green in the "think" section.

5. Experiments

First, we introduce three baseline machine learning methods: Support Vector Machine (SVM), Random Forest (RF), and Multi-Layer Perceptron (MLP). Next, we present the evaluation metrics used in this study, followed by the configuration and optimization details of the machine learning baselines, RAG, SFT, and RL. Subsequently, we analyze and discuss the experimental results.

5.1. Conventional Machine Learning Baselines

To provide a controlled reference under the same data split, we implement three conventional machine learning baselines: SVM, RF, and MLP. These models are trained and evaluated using the same training and test partitions as the proposed LLM-based configurations.

5.1.1. Feature Engineering

Since conventional machine learning models are not capable of directly processing raw textual data, we construct a feature engineering pipeline to convert text into numerical representations.

We use dense semantic representations derived from a pre-trained embedding model. Specifically, each fault description is encoded using the Qwen3-Embedding-0.6B [54] model to produce a fixed-dimensional dense vector. The resulting embeddings are further normalized using L_2 normalization to stabilize the feature scale across samples. This design allows conventional classifiers to operate directly in a semantically meaningful representation space, reducing the dependence on manual feature engineering.

5.1.2. Label Processing

To reduce inconsistency in textual labels, we normalize the cause and action labels through a clustering-based approach. Specifically, labels in the training set are first encoded into embeddings, and HDBSCAN [56] is applied to group semantically similar expressions. Each cluster is then represented by a canonical label, forming a unified label space. The clustering process is performed only on the training set to avoid data leakage.

5.2. Evaluation Metrics

All models, including traditional machine learning and LLM-based methods, were evaluated under a unified evaluation protocol.

Unlike conventional classification tasks, text-based fault diagnosis does not have a unique ground-truth expression. A single fault may correspond to multiple semantically equivalent descriptions of causes and maintenance actions. Therefore, exact string matching is not an appropriate evaluation criterion. Instead, we adopt a semantic similarity-based evaluation framework, where model predictions are compared with reference answers in the embedding space. This allows semantically equivalent but lexically different responses to be treated as correct, thereby providing a more realistic assessment of model performance in practical applications.

In addition to model-based evaluation metrics, we also conducted human-based evaluation to assess the practical usefulness and reliability of the generated diagnostic results.

5.2.1. Model-Based Evaluation

We implemented a model-based semantic evaluation framework to ensure a direct and fair comparison between different models. The predicted labels from SVM, RF, and MLP were first mapped back to their corresponding textual descriptions of causes and actions. This transformation enables a consistent comparison with the free-form textual outputs generated by LLMs.

For each prediction, we consider only the top five candidates in the predicted causes and actions. Each candidate is compared with the ground truth, and the one with the highest similarity score is retained. This design aligns with the multi-candidate nature of fault diagnosis and ensures consistency across model types.

Let $\{y_i\}_{i=1}^{N}$ denote the generated predictions and $\{\hat{y}_i\}_{i=1}^{N}$ the corresponding ground-truth responses. For each instance i, we decompose y_i and $\hat{y}_i$ into the *cause* part $y_{c,i}, \hat{y}_{c,i}$ and the *action* part $y_{a,i}, \hat{y}_{a,i}$, and measure their quality via cosine similarity in the embedding space:

$$\mathrm{Sim}_{v,i} = \cos\big(f_{\mathrm{emb}}(y_{v,i}),\, f_{\mathrm{emb}}(\hat{y}_{v,i})\big), \quad v \in \{c, a\}, \tag{7}$$

where $f_{\mathrm{emb}}(\cdot)$ is the embedding function and $\cos(\cdot)$ denotes cosine similarity.

The embedding model used here is Qwen3-Embedding-0.6B [54]. It should be noted that this embedding model is different from the scoring model employed in Equation (6) during reinforcement learning, in order to ensure an unbiased evaluation.

The overall model-based evaluation score is then defined as

$$\text{Score}_{\text{model}} = \frac{1}{2N} \sum_{i=1}^{N} \left(\text{Sim}_{c,i} + \text{Sim}_{a,i} \right). \tag{8}$$

We further present an accuracy measure that reflects whether the model predictions surpass a similarity threshold. Specifically, a response is considered correct if and only if both the cause and the action achieve cosine similarity values greater than a predefined threshold τ. Formally,

$$\text{Accuracy} = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}\{\text{Sim}_{c,i} > \tau\} \cdot \mathbb{1}\{\text{Sim}_{a,i} > \tau\}, \tag{9}$$

where $\mathbb{1}$ denotes the indicator function. The similarity threshold τ is selected from {0.6, 0.7, 0.8} based on validation results, with $\tau = 0.7$ offering a reasonable trade-off between strict matching and semantic tolerance. This threshold-based accuracy is introduced to provide an intuitive and interpretable metric, indicating whether the generated response reaches a practically acceptable level of semantic alignment with the reference answer.

5.2.2. Human-Based Evaluation

To complement the model-based evaluation, we further conducted a human-based assessment to evaluate the practical reliability and usefulness of the generated diagnostic results. The human evaluation was conducted by HVAC maintenance experts from Ningbo Rail Transit. The human-based evaluation score is defined as

$$\text{Score}_{\text{human}} = \frac{1}{5NM} \sum_{i=1}^{N} \sum_{m=1}^{M} R_{m,i}, \tag{10}$$

where M is the number of human evaluators and $R_{m,i}$ denotes the raw score assigned by the m-th domain expert for the i-th output. Experts assigned a score R from 0 to 5 for each response based on both the correctness and practical value of the proposed reasoning and solutions. A higher score indicates better performance, and the detailed scoring criteria are provided in Appendix A.

We set $M = 20$ and $N = 20$, meaning that for each model, 20 experts evaluated 20 samples. To ensure the reliability and interpretability of this evaluation, the samples were randomly drawn from the test set. To avoid potential bias, the experts were blinded to the model that generated each response, and the responses from different models were randomly ordered before scoring. Each expert scored the responses independently.

To assess the inter-rater reliability of the expert ratings, we computed the intraclass correlation coefficient (ICC) under a two-way random-effects model with absolute agreement, averaged over these expert raters [57].

5.3. Detailed Settings

The large language model employed in this study was Qwen3-0.6B [11], via the `HuggingFace` package [58], configured as summarized in Table 1 to achieve a balance between generation diversity and response stability. The thinking mode was consistently enabled, allowing the model to explicitly generate intermediate diagnostic reasoning steps before producing the final response. All experiments were conducted on a workstation equipped with an Intel Xeon W-1270E CPU and an NVIDIA RTX A4000 GPU.

Table 1. Inference parameters of the Qwen3 model. Temperature controls output randomness; Top-p selects the minimal token set; Top-k limits token candidates; Min-p sets the minimum token probability.

Temperature	Top-p	Top-k	Min-p
0.6	0.95	20	0

Machine Learning: For SVM, we adopted a Radial Basis Function (RBF) kernel with class balancing enabled. For RF, a forest with 500 decision trees was used without additional feature scaling. For MLP, a three-layer neural network with hidden-layer dimensions of (512, 256, 128) was employed, and it was trained using the Adam optimizer with adaptive learning rate scheduling.

Retrieval-Augmented Generation: We employed an embedding model based on CoSENT [59], implemented via the `text2vec` package [60], to compute semantic similarity between fault descriptions. A similarity threshold was introduced to filter low-quality retrievals, and the threshold was tuned within the range of $[0.7, 0.9]$ based on validation experiments, with 0.8 achieving strong retrieval precision while maintaining sufficient coverage. Each query retrieved the top-k similar records, where k was selected from $\{5, 10, 15\}$ based on validation performance, and $k = 10$ effectively limited irrelevant retrievals and preserved informative samples. The retrieved cases were then structured and incorporated into the model input.

Supervised Fine-Tuning: We employed a full-parameter fine-tuning strategy and applied a linear warm-up over the first 10 steps to stabilize the initial gradient updates. The learning rate was selected from $\{1 \times 10^{-5}, 2 \times 10^{-5}, 5 \times 10^{-5}\}$ based on validation performance, where 2×10^{-5} yields stable convergence and optimal accuracy. We used a total of 1467 augmented training samples, corresponding to 264 steps over three epochs. All other hyperparameters followed the default settings in HuggingFace.

Reinforcement Learning: The RL hyperparameters were selected based on validation experiments and prior studies on policy optimization. During GRPO training, we set the total number of steps to 10,000, with a per-device batch size of 1 and 8 gradient accumulation steps, resulting in an effective batch size of 8. The learning rate was fixed at 1×10^{-5} with 50 warm-up steps. We generated 8 samples per input and set the maximum completion length to 1024 tokens. The KL penalty coefficient β was set to 0.1. For the reward function, the weights were chosen from a candidate set to reflect their relative importance, where semantic accuracy was prioritized ($\alpha_3 = 1$), while format and content rewards were assigned smaller weights ($\alpha_1 = \alpha_2 = 0.2$). This configuration was validated to achieve the best empirical performance. Semantic evaluation was conducted by the Qwen3-Reranker-0.6B model [54] with $t = 0.7$ in Equation (6), offering a reasonable trade-off between strict matching and semantic tolerance.

5.4. Results

The proposed framework follows a staged paradigm of retrieval, distillation, and optimization. Specifically, RAG enriches sparse maintenance data, SFT distills this knowledge into model parameters, and GRPO-based RL refines diagnostic reasoning. The experimental configurations reported below correspond to progressively enabling these stages and, thus, provide a scalable evaluation of the contribution of each module.

Table 2 and Figure 8 present a comprehensive comparison of prediction results on the test set, covering traditional machine learning baselines (SVM, RF, and MLP) alongside four LLM-based configurations: the baseline Qwen3-0.6B model, the RAG-enhanced model, the cold-start model with SFT, and the RL model optimized with GRPO. In addition, all

LLM-based methods were further evaluated using human-based assessments to measure their reasoning quality and practical utility.

Table 2. Comparison of model- and human-based evaluation metrics for four large language models in HVAC fault diagnosis.

		SVM	RF	MLP	Qwen3	+RAG	+SFT	+SFT+RL
All	Accuracy	0.272	0.341	0.366	0.323	0.554	0.448	0.635
	$\text{Score}_{\text{model}}$	0.686	0.705	0.714	0.698	0.746	0.727	0.768
	$\text{Score}_{\text{human}}$	/	/	/	0.509	0.723	0.698	0.754
	$\text{ICC}(A,k)$	/	/	/	0.502	0.524	0.967	0.956
Cause	Accuracy	0.422	0.485	0.513	0.516	0.668	0.615	0.704
	$\text{Score}_{\text{model}}$	0.683	0.698	0.710	0.698	0.743	0.733	0.756
Action	Accuracy	0.451	0.518	0.549	0.495	0.724	0.582	0.839
	$\text{Score}_{\text{model}}$	0.690	0.713	0.719	0.700	0.749	0.722	0.780

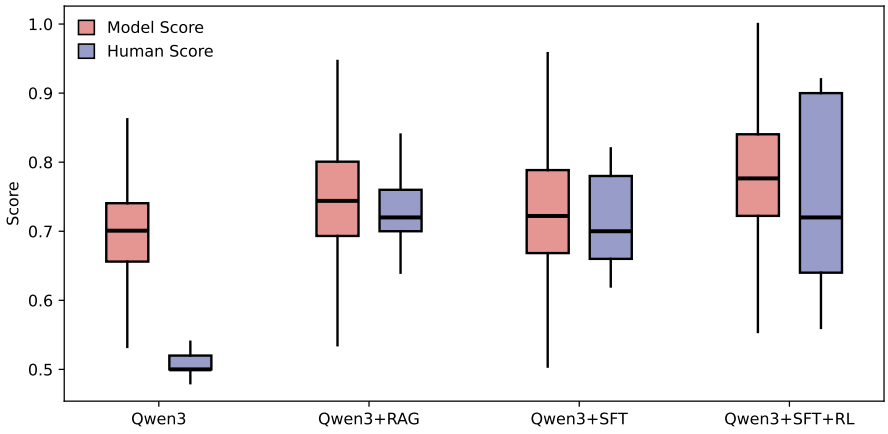


Figure 8. Boxplots of model- and human-based evaluations for large language models. The bold line within each box indicates the median score, the box edges denote the interquartile range, and the whiskers show the overall score distribution. Higher values correspond to better model performance.

From Table 2, traditional machine learning methods show relatively low accuracy, with SVM, RF, and MLP obtaining accuracies of 0.272, 0.341, and 0.366, respectively. The Qwen3 model without post-training shows accuracy comparable to these traditional approaches. In contrast, all post-trained LLMs consistently outperform the baselines mentioned above. These results suggest the necessity of using advanced language models for text-based fault diagnosis tasks and highlight the importance of post-training adaptation for achieving better performance.

The proposed multi-stage post-training model achieves the best performance in Table 2, with an average model-based score of 0.768 and an accuracy of 0.635, representing improvements of approximately 0.1 and 0.3, respectively, over the baseline Qwen3-0.6B model. The RAG-enhanced model also improves accuracy over the Qwen3 baseline, indicating that incorporating historical data retrieval is effective. We also observed that the cold-start model trained only with SFT does not surpass the RAG-enhanced model in Table 2, indicating that SFT alone is insufficient to provide the model with effective reasoning capabilities in this task. This degraded performance can be attributed to the fact that RAG dynamically injects instance-specific contextual knowledge during inference, whereas SFT attempts to statically compress this knowledge into model parameters, which can be particularly challenging under limited and noisy training data. The proposed Qwen3+SFT+RL model, which does not use retrieval at inference, still outperforms the RAG model that relies on inference-time

retrieval. This suggests that the observed gains stem from knowledge internalization and task-specific RL optimization rather than from retrieval availability alone.

Furthermore, consistent trends are observed in the segmented results of cause identification and action recommendation in Table 2, where the RL-trained model achieves the best results, with particularly notable improvements in the action part. These findings suggest that the proposed multi-stage post-training framework enhances the model's ability to generate practical maintenance recommendations.

As can be seen from Figure 8, the proposed multi-stage post-training model achieves higher scores in both model and human evaluations, although the variance in human-based score is relatively large. To examine whether this variation reflects inconsistency among the experts, we compute the inter-rater reliability ICC(A, k) for each model in Table 2. The resulting ICC values are 0.502, 0.524, 0.967, and 0.956 for the original Qwen3, +RAG, +SFT, and +SFT+RL models, respectively, indicating moderate agreement for the original and +RAG models and good agreement for the +SFT and +SFT+RL models [61]. The high agreement for the post-trained models suggests that their relatively large score spread mainly reflects differences in sample difficulty rather than inconsistency among the experts, thereby supporting the reliability of the human-based evaluation. In contrast, the lower agreement for the original and RAG models is consistent with their less structured outputs, for which expert judgments tend to diverge more.

In summary, RAG improves the baseline by using similar historical cases, while SFT alone remains insufficient. The multi-stage framework enables LLMs to achieve higher diagnostic accuracy and deliver more effective maintenance guidance.

5.5. Discussion

Our LLM-based approach focuses on processing maintenance records derived from unstructured textual data, whereas traditional machine learning methods predominantly depend on sensor signals and structured measurement data. As shown in Table 2, although these baselines provide a useful reference, their performance is lower than that of LLM-based methods. In addition, they formulate the task as a closed-set classification problem with a predefined label space. Therefore, they lack the ability to handle unseen fault types, which limits their applicability in real-world tasks.

The observed performance gains stem from the complementary roles of RAG, SFT, and RL. RAG compensates for the brevity of fault descriptions by injecting contextual priors from similar historical cases. SFT then provides the model with the structured diagnostic patterns. RL further refines the model by aligning its reasoning with the diagnostic CoTs through the semantic reward. The combination of these stages enables the model to both learn domain knowledge and produce structured diagnostic reasoning, leading to the substantial improvement over the baseline.

One notable finding is that our approach outperforms the RAG-based method, although the training data are constructed from RAG retrieval results. RAG improves diagnosis by providing relevant historical cases at inference time, while SFT learns directly from the augmented supervision and may inherit its noise. RL, in contrast, uses task-specific reward feedback to favor responses that better match the reference diagnosis and maintenance action, making it less sensitive to noise in the augmented data. However, since the current semantic reward measures alignment with reference answers rather than true reasoning quality or operational safety, future work will employ reasoning-focused benchmarks and expert assessment to evaluate diagnostic reasoning, alongside safety-oriented metrics such as fault-cause recall, hallucination rate, unsafe-action rate, and expert-confirmed action correctness before real deployment.

It should also be noted that, due to computational constraints, this study investigated the effectiveness of the proposed algorithm only on the 0.6B-parameter model, while its performance on larger models remains unexplored. According to the scaling law, model performance generally improves as the model size increases. In the future, we plan to extend our experiments to larger models once sufficient computational resources become available.

A further limitation is that, although the proposed method has been validated as effective on our own dataset, it remains unevaluated against stronger NLP baselines and recent LLM-based fault-diagnosis approaches. Since those studies are evaluated on different datasets, task formulations, and evaluation metrics, their reported results are not directly comparable with ours. Such comparisons, together with cross-dataset and cross-domain evaluation, are therefore important next steps before making broader claims about general-purpose fault-diagnosis performance.

Moreover, the reasoning capability of our model primarily relies on historical patterns observed during training. Consequently, the diagnostic performance degrades when encountering fault types that are absent from the training history. In addition, RAG-generated Chain-of-Thought reasoning may contain incorrect causal relationships, which can introduce noise into the augmented training data. Future work will evaluate the quality of the generated reasoning through quantitative measures and expert validation, and further augment the training corpus with domain knowledge from textbooks and technical manuals to improve generalization to rare or unseen fault cases.

Another characteristic of the real-world dataset is its long-tailed distribution. In this work, the retrieval-based design helps the framework handle rare-but-seen faults by retrieving similar historical cases. Nevertheless, performance may still decline for completely unseen fault types, as the experiments indicate that the untrained Qwen3 performs poorly on this task. We also adopted a chronological split to mirror real deployment. While cross-line, cross-station, or cross-equipment validation would be desirable, such experiments currently yield too few test samples for reliable estimates. Therefore, a systematic comparison between frequent and infrequent faults, along with broader generalization across lines, stations, and equipment types, remains for future work.

6. Conclusions

This study proposes a post-training framework for accurate fault diagnosis in urban rail transit HVAC systems. It builds upon a large language model (LLM) by incorporating domain-specific data augmentation and a post-training strategy consisting of supervised fine-tuning and reinforcement learning. The approach is grounded on a complex, real-world dataset constructed from textual maintenance records of Ningbo Rail Transit, encompassing 33 equipment categories and over 100 distinct fault types.

The experimental results show that the framework significantly improves the diagnostic accuracy and reasoning capability of the baseline model. The trained model produces a clear diagnostic chain of thought and achieves a 97% improvement in diagnosis accuracy compared to the original untrained model on the test dataset, while maintaining relatively low computational cost compared to large-scale models, suggesting the potential for real-time deployment. Overall, this post-training approach effectively adapts a pre-trained LLM for accurate and efficient HVAC fault diagnosis and provides a practical strategy for deploying lightweight, specialized models in industrial settings.

In practical applications, the proposed model can be integrated into multiple operational workflows within metro equipment maintenance. It can serve as an intelligent assistant in the maintenance ticketing system by automatically interpreting newly submitted fault descriptions and providing preliminary diagnostic insights to help on-site inspections. It can also support post-event analysis by generating standardized summaries

of fault causes and maintenance actions, thereby improving the consistency and completeness of maintenance documentation across stations. In addition, the model offers value for staff training, particularly for newly recruited technicians, as it provides examples of typical diagnostic thinking and helps them become familiar with a wide range of fault scenarios more quickly.

In future work, the proposed post-training framework could be extended to other electromechanical systems in urban rail transit, such as traction systems, lighting systems, and escalators. Investigating larger or more efficient model architectures may also help improve diagnostic accuracy and robustness. Furthermore, integrating the LLM with existing operation and maintenance systems could enable real-time access to sensor data, historical logs, and control interfaces. This type of integration would support the development of an intelligent agent for proactive monitoring and tool-augmented reasoning, thereby facilitating its deployment in the real world.

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Abbreviations

The following abbreviations are used in this manuscript:

AC	Air Conditioning
AHU	Air Handling Unit
CoT	Chain of Thought
GRPO	Group Relative Policy Optimization
HVAC	Heating, Ventilation, and Air Conditioning
ICC	Intraclass Correlation Coefficient
LLM	Large Language Model
LoRA	Low-Rank Adaptation
MLP	Multi-Layer Perceptron
RAG	Retrieval-Augmented Generation
RBF	Radial Basis Function
RF	Random Forest
RL	Reinforcement Learning

SFT	Supervised Fine-Tuning
SVM	Support Vector Machine
URT	Urban Rail Transit

Appendix A. Human Evaluation Criteria

The expert questionnaire adopts a six-level rating scale (0–5) to assess the quality of the model-generated fault cause analyses and recommended actions. The evaluation criteria are defined as follows:

- 0— Completely incorrect or vacuous response with no reference value;
- 1—Mostly incorrect, with only minor relevant content;
- 2—Partially correct but vague, with limited maintenance utility;
- 3—Largely correct with minor errors or omissions, offering some maintenance guidance;
- 4—Highly accurate with minimal errors, providing substantial practical guidance;
- 5—Fully accurate and comprehensive, capable of directly guiding on-site maintenance.

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